Conic Sections at a Glance

My Pre-Calculus reviewer. A conic section is the curve you get when a plane slices through a double cone (two cones joined tip to tip). Depending on the angle of the cut, you get one of four curves: a circle, a parabola, an ellipse or a hyperbola.

Circle

A circle is the set of all points in a plane that are the same distance, called the radius r, from a fixed point called the center.

- Standard form with center (h, k): (x - h)^2 + (y - k)^2 = r^2
- Center at the origin: x^2 + y^2 = r^2

You get a circle when the plane cuts perpendicular to the cone's axis.

Parabola

A parabola is the set of points that are equally far from a fixed point (the focus) and a fixed line (the directrix). The vertex is the point exactly halfway between the focus and the directrix, and p is the directed distance from the vertex to the focus.

- Opens up or down, vertex (h, k): (x - h)^2 = 4p(y - k). It opens up if p > 0 and down if p < 0.
- Opens right or left, vertex (h, k): (y - k)^2 = 4p(x - h). It opens right if p > 0 and left if p < 0.
- For (x - h)^2 = 4p(y - k), the focus is (h, k + p) and the directrix is y = k - p.

The length of the latus rectum, the chord through the focus perpendicular to the axis, is |4p|.

Ellipse

An ellipse is the set of points where the sum of the distances to two fixed points (the foci) is constant. That constant equals 2a, the length of the major axis.

- Horizontal major axis, center (h, k): (x - h)^2/a^2 + (y - k)^2/b^2 = 1, with a > b
- Vertical major axis: (x - h)^2/b^2 + (y - k)^2/a^2 = 1, with a > b
- Relationship: c^2 = a^2 - b^2, where c is the distance from the center to each focus

The vertices are a units from the center along the major axis, and the co-vertices are b units along the minor axis. A quick tip: the larger denominator tells you the direction of the major axis.

Hyperbola

A hyperbola is the set of points where the absolute value of the difference of the distances to two foci is constant, equal to 2a. It has two separate branches.

- Opens left and right, center (h, k): (x - h)^2/a^2 - (y - k)^2/b^2 = 1
- Opens up and down: (y - k)^2/a^2 - (x - h)^2/b^2 = 1
- Relationship: c^2 = a^2 + b^2
- Asymptotes for a hyperbola centered at the origin opening left and right: y = (b/a)x and y = -(b/a)x

For a hyperbola, a^2 always sits under the positive term, and a need not be bigger than b.

Telling them apart from the general equation

Many problems give the equation in general form, Ax^2 + Cy^2 + Dx + Ey + F = 0 (no xy term). Just look at A and C, the coefficients of x^2 and y^2:

- Circle: A = C (same sign and same value)
- Parabola: either A = 0 or C = 0, but not both (only one variable is squared)
- Ellipse: A and C have the same sign but different values (AC > 0, A not equal to C)
- Hyperbola: A and C have opposite signs (AC < 0)

These tests assume the graph is not degenerate (a point, lines, or nothing).

Another clue is eccentricity, e, which measures how stretched a conic is. A circle has e = 0, an ellipse has 0 < e < 1, a parabola has e = 1, and a hyperbola has e > 1. For ellipses and hyperbolas, e = c/a.

Worked example

Identify the conic and find its key parts: x^2 + y^2 - 6x + 4y - 12 = 0.

Step 1. Classify. A = 1 and C = 1, so A = C. It is a circle.

Step 2. Group terms and move the constant: (x^2 - 6x) + (y^2 + 4y) = 12.

Step 3. Complete the square. Half of -6 is -3, and (-3)^2 = 9. Half of 4 is 2, and 2^2 = 4. Add 9 and 4 to both sides:

(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4

Step 4. Factor: (x - 3)^2 + (y + 2)^2 = 25.

Step 5. Read off the answer. The center is (3, -2) and the radius is sqrt(25) = 5.

Quick check

- Q: What is the standard equation of a circle with center (h, k) and radius r? A: (x - h)^2 + (y - k)^2 = r^2.
- Q: Which conic has c^2 = a^2 + b^2? A: The hyperbola.
- Q: In general form, what does it mean if A and C have opposite signs? A: The conic is a hyperbola.
- Q: What is the eccentricity of every parabola? A: Exactly 1.
- Q: Where is the focus of x^2 = 8y? A: 4p = 8, so p = 2 and the focus is (0, 2).